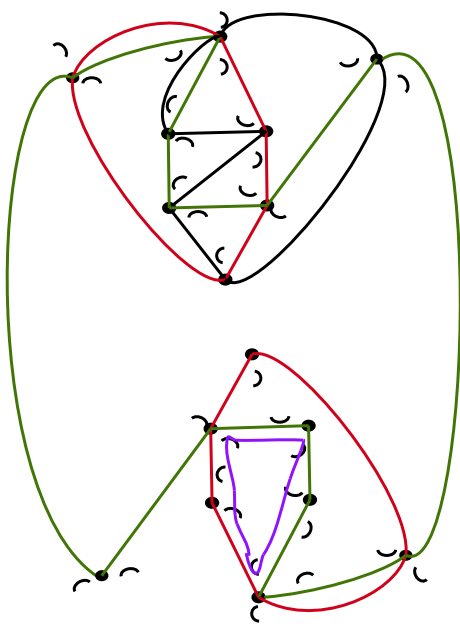
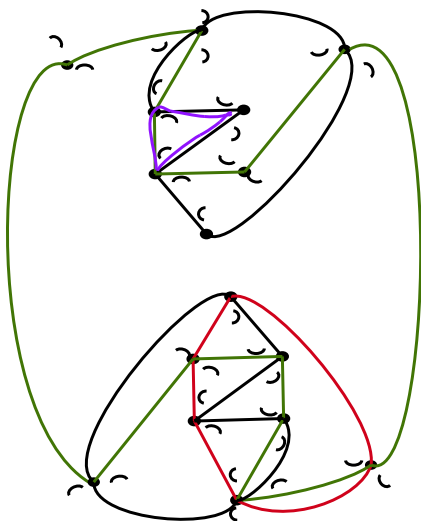




SOME OBSERVATIONS

The choice of operation 1 removes the compatible triangles occurred due to operation 2 and 3.

Almost in all cases deletion of compatible cycle leads to triangles or cycles of length 5 rendering the Eulerian circuit reducible, with a few exceptions.

The compatible cycles obtained from the cycle decomposition of $\backslash\mathit{math-container}$ are not vertex-disjoint to each other in one half of $\backslash\mathit{math-container}$.

So any such compatible cycle belonging to $\backslash\mathit{math-container}$ that does not give a $\backslash\mathit{math-container}$ after being removed in one half of $\backslash\mathit{math-container}$ will not give a $\backslash\mathit{math-container}$ when removed in both halves of $\backslash\mathit{math-container}$ produce same reducible circuit in the respective halves of $\backslash\mathit{math-container}$.

Also, if we remove a compatible cycle we get from $\backslash\mathit{math-container}$ and find another compatible cycle that did not occur in $\backslash\mathit{math-container}$ (if vertex disjoint), and remove its edges it will leave $\backslash\mathit{math-container}$ disconnected.

Case with multiple compatible-cycle deletion:

There are only two cycles of length 2 in $\backslash\mathit{math-container}$ and they don't produce a $\backslash\mathit{math-container}$, whether removed individually or with other vertex-disjoint compatible cycles.

This compatible 2-cycle is also in $\backslash\mathit{math-container}$ but does not become a part of the cycle decomposition. There are no other compatible 2-cycles.

Given all these observations there are no more possible compatible cycles to check